



FPVspeed™

DIY Module Series

# FPV Video Receiver

**FS2540RX**

2500~4000MHz -93±2dBm SPI &amp; RSSI

## 1. 通用规格 (General)

| FS2540RX                    |                      |
|-----------------------------|----------------------|
| Weight                      | 5.5g±0.1g            |
| Dimensions                  | L37 x W26.3 x H4.5mm |
| DC Characteristics          |                      |
| Power Supply                | 5.0V                 |
| Current Consumption         | 380mA ±30mA @5V      |
| Environmental Specification |                      |
| Operating Temperature       | -10~+65°C            |
| Storing Temperature         | -30~+85°C            |
| Operating Humidity          | 85%RH                |
| RF                          |                      |
| Receiving Frequency Range   | 2500~4000MHz         |
| Voltage Standing Wave Ratio | 2:1                  |
| Demodulation System         | FM/PLL               |
| IF                          | 480MHz               |
| ANT. Input Impedance        | 50Ω, Typ.            |
| LO Frequency Stabilization  | ±200kHz              |
| LO Frequency Precision      | ±200kHz              |
| LO Control                  | PLL                  |
| Input LO Leak               | -65dBm               |
| Receiving Sensitivity       | -93dBm±2dBm          |
| Input Level Range           | -93 dBm ~+5 dBm      |
| Video Characteristics       |                      |
| Video Output Impedance      | 75Ω, Typ.            |
| Video Output Level          | 1±0.2Vp-p, Typ.      |
| Video Polarity              | Negative             |
| Video Frequency Response    | ±5dB, Max. 50Hz~6MHz |
| Differential Gain           | ±5%, Max             |
| Differential Phase          | ±5Deg., Max          |
| 3Db IF WIDEBAND             | 16.5MHz              |
| S/N                         | 38dB, Min            |



## Applications

- FPV Racing Competition
- Wireless AV Transmission System
- Security Surveillance Communication
- Scientific Research and Education

## 2. 测试条件 (Test Condition)

| Ambient temperature | 25°C               |
|---------------------|--------------------|
| Video input voltage | 1.0Vp-p, 150 (IRE) |



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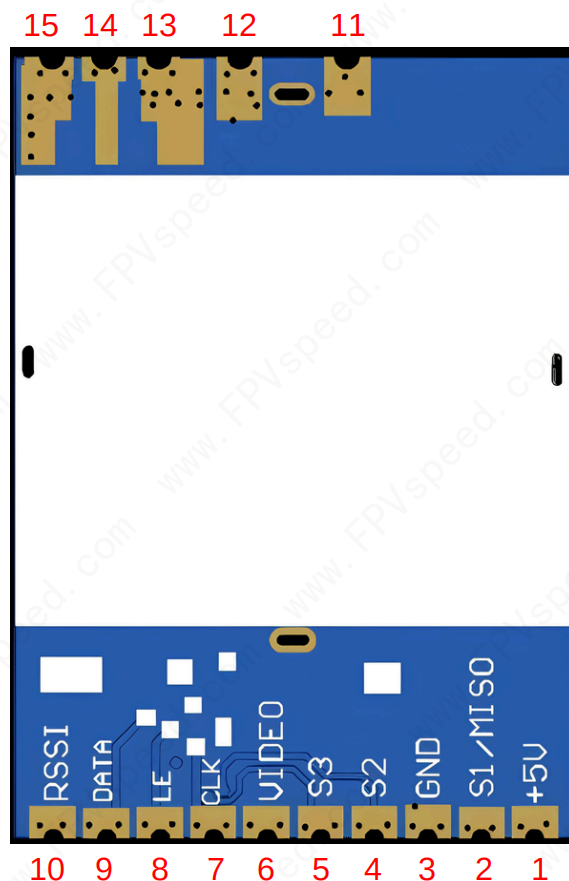
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## 3.引脚分布 (Pin Assignment)

| Pin No. | Function  |
|---------|-----------|
| 1       | +5V IN    |
| 2       | S1 / MISO |
| 3       | GND       |
| 4       | S2 / NC   |
| 5       | S3 / NC   |
| 6       | Video Out |
| 7       | SPI_CLK   |
| 8       | SPI_LE    |
| 9       | SPI_DATA  |
| 10      | RSSI      |
| 11      | GND       |
| 12      | GND       |
| 13      | GND       |
| 14      | ANT       |
| 15      | GND       |



- The **FS2540RX** is a module that supports CVBS signal output. It features low latency and strong anti-interference capabilities, and is widely used in fields such as FPV (First-Person View) drone flight, outdoor entertainment, racing competitions, and professional aerial photography. It allows developers to conduct functional expansion and customized development based on the existing SPI protocol.

**Supports SPI frequency mode, 2500–4000 MHz**



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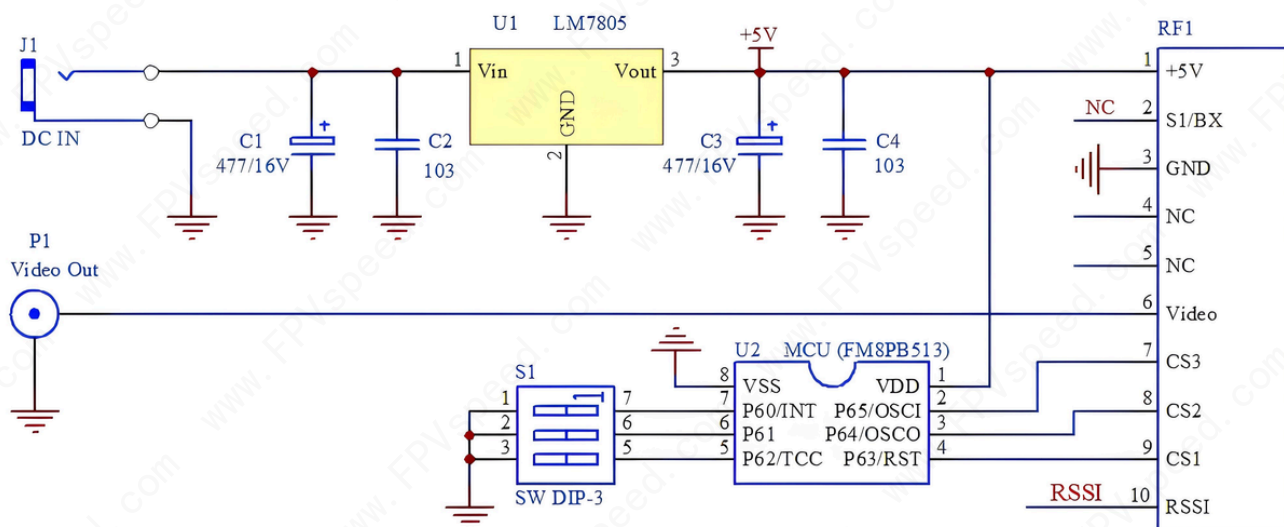
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## 4.应用电路 (Application Circuit)



## 5.结构尺寸 (Dimension)

